

<u>CSXXX</u>	<u>Computer Vision (CV)</u>
Credits: 3 (3-0-0)	
Pre-requisite for this course:	• Data structures • Linear algebra and vector calculus • Machine learning fundamentals • Programming knowledge (C++ or Python) • Opensource libraries (computer vision libraries, image processing libraries)
Course Objective:	The course objectives: • Introduction of the image fundamentals • Developing mathematical intuition and foundation of computer vision principles • Applications of machine learning in computer vision to solve real-world problems
Course Outcome:	On completion of the course, a student should be able to: • Detailed understanding of image filtering, transformations, geometric and physical models of image formation • In-depth understanding of computer vision principles • Practical exposure and relevance of machine learning techniques in computer vision

Syllabus

Module I [10] Hours	Image processing fundamentals: Introduction and Overview. Image Formation: Geometric primitives and transformations, photometric image formation, the principle of digital camera. Image Processing: Image transformations, image filtering, edge detection, corner detection.
Module II [10] Hours	Geometrical techniques in computer vision: Feature Detection and Matching: Feature detectors, feature descriptors, feature matching and feature tracking. Camera geometry: Camera projections, camera calibration, camera pose estimation. Depth Estimation: Epipolar geometry, sparse and dense correspondence, multi-view stereo. Two-view Structure from Motion, Object tracking.
Module III [10] Hours	Deep learning for computer vision: Deep neural networks, convolutional neural networks (CNN), Three-dimensional CNNs, Recurrent neural networks, Transformers, and Generative models and applications.
Suggested Books:	1. Computer Vision: Algorithms and Applications, Springer London, (2011). Richard Szeliski. 2. Multiple View Geometry in Computer Vision, Cambridge University Press, (2004). Richard Hartley, Andrew Zisserman. 3. Computer Vision: A Modern Approach, Pearson, (2012). David Forsyth, Jean Ponce.
Evaluation:	7. Quizzes: 15% 8. Mid Term: 30% 9. End Term Exam: 50% 10. Teacher's Assessment: 5%